

Information Retrieval and Text Mining

Evaluation of Information Retrieval systems

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Session outline

1. Evaluation measures in IR

- Precision, Recall
- Precision/Recall curve
- F_β Score
- F_1 Score
- Confusion Matrix
- ROC curve, AUC
- Other IR evaluation measures

Learning outcomes

At the end of this session we will be able to:

- Compute and explain precision, recall, F_β score and other evaluation measures
- Represent and analyze Precision/Recall curves
- Represent and analyze ROC curves and AUC
- Design IR evaluation protocols, define proper indicators/measures

1. Evaluation of IR systems

A general overview of the IR assessment: precision, recall, F_β and F_1 score.

Graphically representing ROC curve, AUC, Precision/Recall curve

Precision and Recall

- Precision, Recall calculation:
 - **Precision** : number of retrieved docs, from all that were obtained, that are relevant to the user's information need

$$\text{Precision} = \frac{\text{Number of Relevant Documents Retrieved}}{\text{Total Number of Documents Retrieved}}$$

- **Recall** : number of relevant docs in collection that are retrieved

$$\text{Recall} = \frac{\text{Number of Relevant Documents Retrieved}}{\text{Total Number of Relevant Documents in the Corpus}}$$

Precision and Recall

- Let us consider the following situation:
 - A group of human experts identified R1, the set of documents that are deemed relevant for a given query Q1: $R1=\{d3, d5, d8, d23, d33, d48, d50, d66, d74, d92\}$
 - Using an automatic retrieval algorithm for the same query Q1 over the same corpus, the ranking obtained was the following:

1	d74	6	d33	11	d101
2	d21	7	d123	12	d90
3	d48	8	d5	13	d92
4	d24	9	d22	14	d100
5	d18	10	d12	15	d1

- The retrieved documents that are relevant to Q1 are represented in bold.

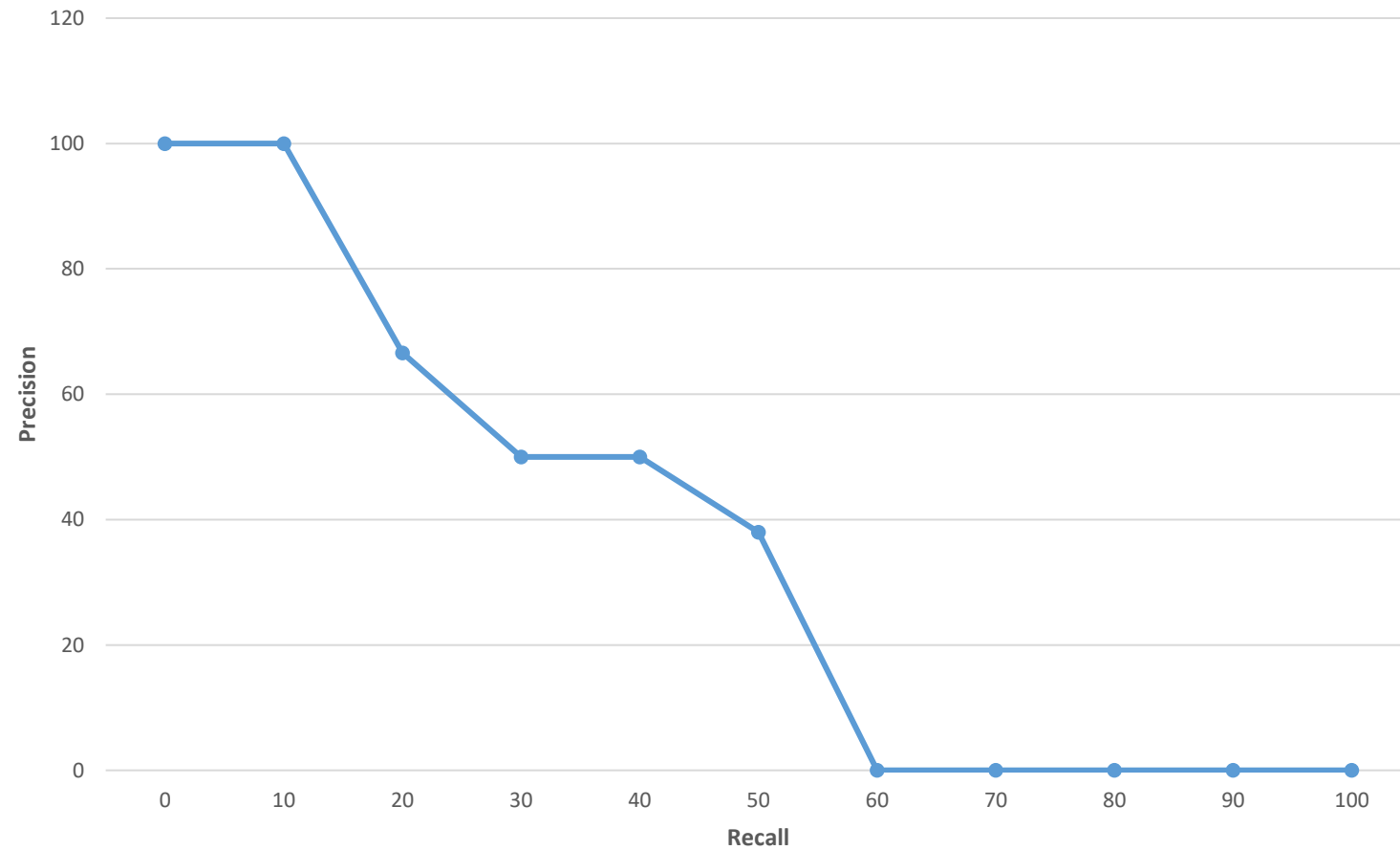
Precision and Recall

1	d74	6	d33	11	d101
2	d21	7	d123	12	d90
3	d48	8	d5	13	d92
4	d24	9	d22	14	d100
5	d18	10	d12	15	d1

- Since the first element of the result set obtained with the retrieval algorithm is relevant ($d74 \in R1$) the precision after assessing the first document retrieved is 1/1 (100%) and the corresponding recall is 1/10 (10%).
- The second relevant element that was retrieved is in the third position of the result set; at this stage, after assessing the top three retrieved documents, we have a precision of 2/3 (66,6%) and a recall of 2/10 (20%)
- The third relevant element is found at the sixth position in the result set; after assessing the first six results retrieved, we have a precision of 3/6 (50%) and a recall of 3/10 (30%)
- What about the precision and recall of the 4th and 5th relevant document?

Precision/Recall curve

R	P
0	100
10	100
20	66,6
30	50
40	50
50	38
60	0
70	0
80	0
90	0
100	0



1.3 F-score

Recall and Precision aggregated.

F_β Score

- **F_β score:**

- Is a metric used in information retrieval to evaluate the performance of ranking algorithms, particularly in relevance feedback scenarios.
- It's an extension of the F_1 score, which combines precision and recall into a single measure.

- $F_\beta = (1 + \beta^2) \times \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}}$ where:

- β is a parameter that controls the relative importance of precision and recall.
- Precision is the ratio of relevant documents retrieved to the total number of documents retrieved.
- Recall is the ratio of relevant documents retrieved to the total number of relevant documents.
- The beta parameter represents the ratio of recall importance to precision importance. $\beta > 1$ gives more weight to recall, while $\beta < 1$ favors precision
- For example, $\beta = 2$ makes recall twice as important as precision, while $\beta = 0.5$ does the opposite. Asymptotically, $\beta \rightarrow +\infty$ considers only recall, and $\beta \rightarrow 0$ only precision.

F₁ Score

- **F₁ score:** Is a metric used to evaluate the performance of a classification model, particularly when dealing with imbalanced classes. It is the harmonic mean of precision and recall, providing a single measure that balances both metrics.
 - It is very used in IR
- $F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ onde

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

1.2 Confusion Matrix

Performance of classification.

Confusion Matrix

- Confusion Matrix : is a table used to evaluate the performance of a classification model.
- It allows visualization of the performance of a model by comparing predicted class labels with true class labels.
- The matrix is particularly useful for assessing the accuracy of a model's predictions.

Confusion Matrix

- How to compute a Confusion Matrix:
 1. You need a test dataset or a validation dataset with expected outcome values
 2. Make a prediction for each row in your test dataset
 3. From the expected outcomes and predictions count:
 1. The number of correct predictions for each class
 2. The number of incorrect predictions for each class, organized by the class that was predicted
- These numbers are then organized into a table, or a matrix as follows:
 - Expected down the side: Each row of the matrix corresponds to a predicted class
 - Predicted across the top: Each column of the matrix corresponds to an actual class

Confusion Matrix

EMAIL	Actual category (labelled by an expert)	Classifier prediction
1	Spam	Spam
2	Non spam	Non spam
3	Non spam	Non spam
4	Non spam	Non spam
5	Non spam	Non spam
6	Spam	Non spam
7	Spam	Non spam
8	Non spam	Non spam
9	Non spam	Spam
10	Spam	Spam

Confusion Matrix

- Example:

	Predicted Non Spam	Predicted Spam
Actual Non Spam	5	1
Actual Spam	2	2

Confusion Matrix

- Example:

	Predicted Spam	Predicted Non Spam
Actual Spam	2	2
Actual Non Spam	1	5

Confusion Matrix

- Example:

	Positive Prediction	Negative Prediction
Positive Class	True (Positive)	False (Negative)
Negative Class	False (Positive)	True (Negative)

Confusion Matrix

- Example:

	Predicted Non Spam	Predicted Spam
Actual Non Spam	5	1
Actual Spam	2	2

F₁ Score

- Example:
 - Suppose that we are trying to classify emails as either spam (positive) or non-spam (negative). After applying a classification algorithm, we obtain the following confusion matrix:

	Predicted Spam	Predicted Non Spam
Actual SPAM	1150	150
Actual Non Spam	200	8500

- True (Positives) (TP) = 1150 (number of correctly classified spam emails)
- False (Positives) (FP) = 200 (number of non-spam emails incorrectly classified as spam)
- False (Negatives) (FN) = 150 (number of spam emails incorrectly classified as non-spam)

F₁ Score

- Example:

	Predicted Spam	Predicted Non Spam
Actual SPAM	1150 (TP)	150 (FN)
Actual Non Spam	200 (FP)	8500 (TN)

- **Precision** = $\frac{1150}{1150+200} = \frac{1150}{1350} \approx 0.852$

- **Recall** = $\frac{1150}{1150+150} = \frac{1150}{1300} \approx 0.885$

- **F1** = $2 \times \frac{0.852 \times 0.885}{0.852 + 0.885} = \frac{1.506}{1.737} \approx 0.866$

1.4 ROC curve

Binary classifier performance analysis.

Receiver Operating Characteristic(ROC) Curve

- Is a graphical plot that summarizes the performance of a binary classification model on the positive class across various threshold settings
- Commonly used in IR to evaluate performance (when results are assessed either as relevant or non-relevant)
- Visualizes the trade-off between True Positive Rate (also known as recall) and False Positive Rate
- The x-axis indicates the False Positive Rate and the y-axis indicates the True Positive Rate.

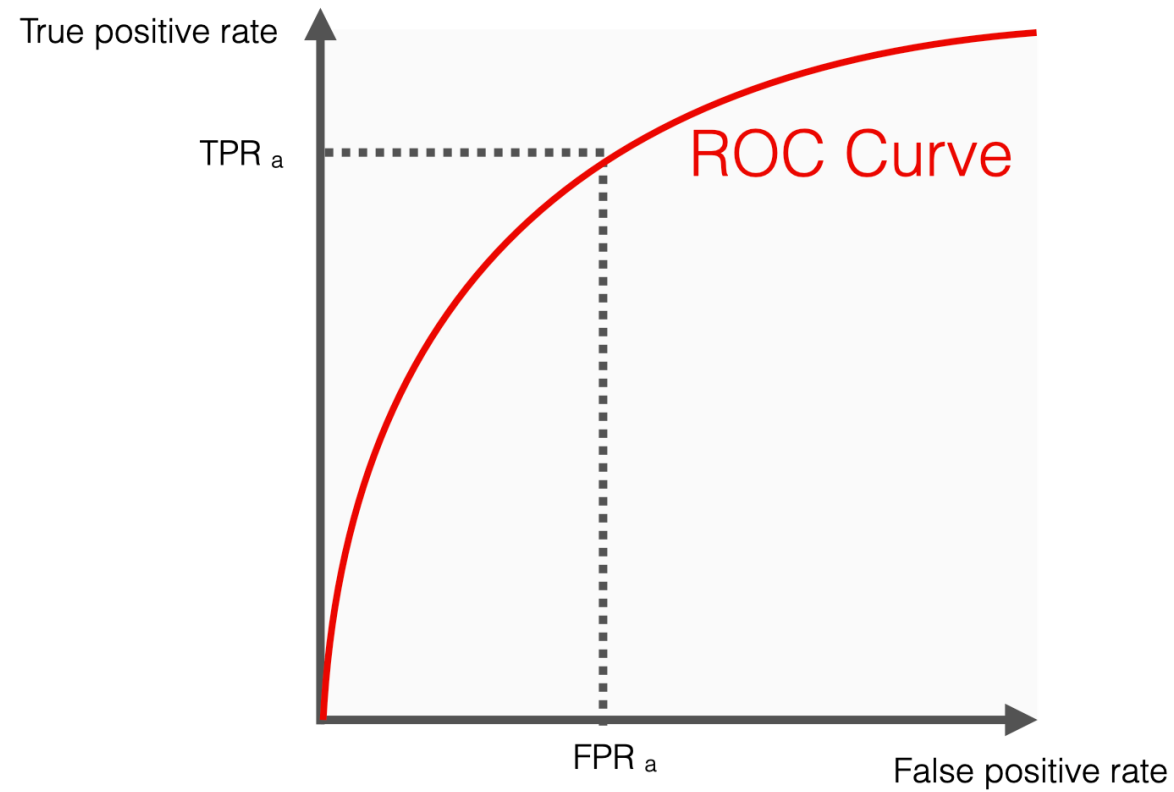
Receiver Operating Characteristic(ROC) Curve

- How to calculate True Positive and False Positive Rates

- $TPR = \frac{TP}{TP + FN}$

- $FPR = \frac{FP}{FP + TN}$

Receiver Operating Characteristic(ROC) Curve



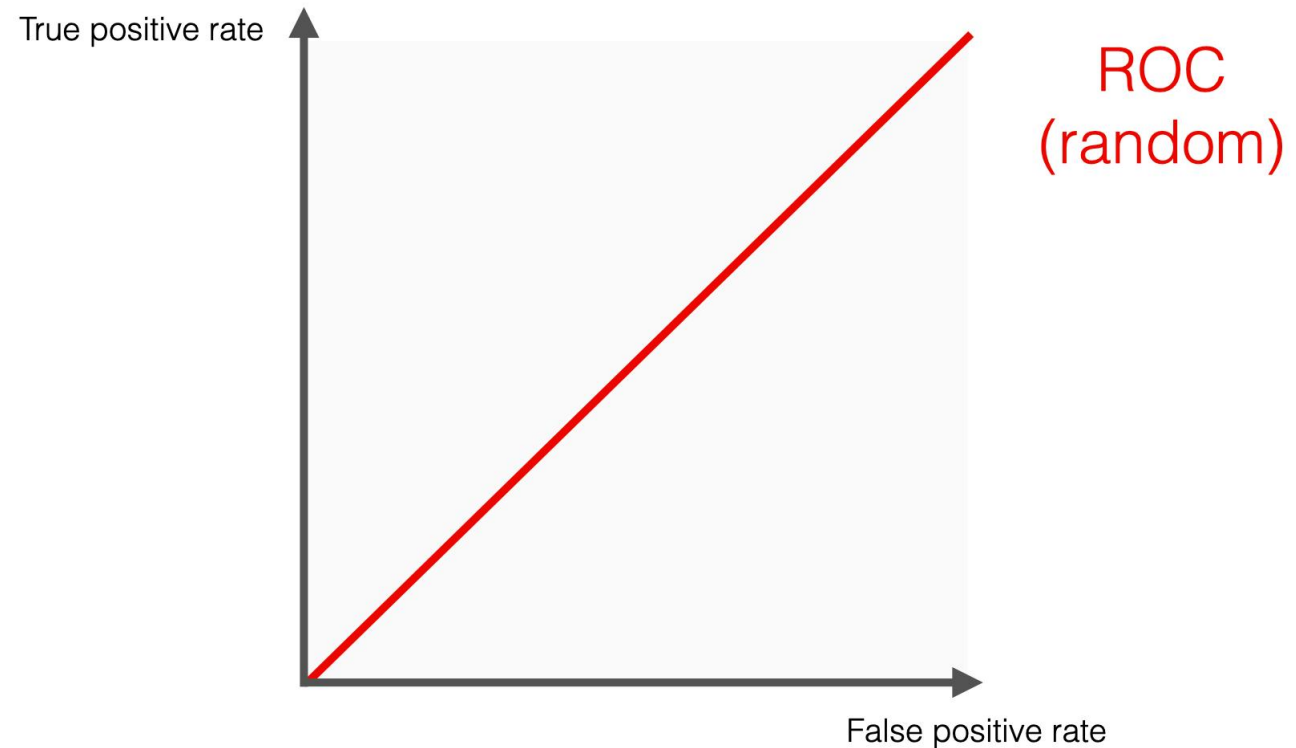
Receiver Operating Characteristic(ROC) Curve

- Perfect scenario



Receiver Operating Characteristic(ROC) Curve

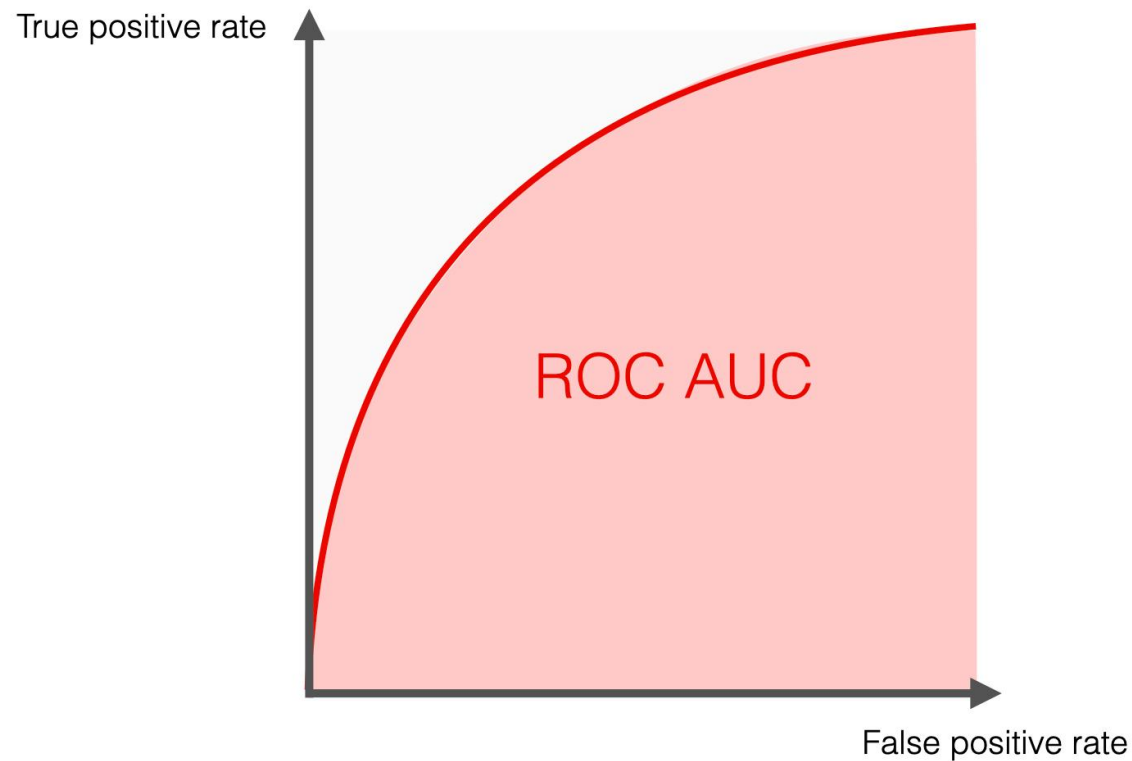
- Worst scenario, Random classifier



ROC Area Under Curve (AUC) Score

- ROC AUC is a score that shows the quality (performance) of the classifier across all possible classification thresholds.
- To get the score, you must measure the area under the ROC curve.
- The ROC AUC score range from 0 to 1, with 0,5 indicating random guessing
- Shows how well the classifier distinguishes positive and negative classes

Receiver Operating Characteristic(ROC) Curve



1.5 Other IR indicators

IR systema performance measures.

Other IR assessment measures

- **Precision at k (P@k)**

Measures relevance in the top results. Useful for web search, where users rarely look past page one.

- **Mean Average Precision (MAP)**

Measures ranking quality across multiple queries. AP is the area under the Precision-Recall curve

- **Normalised Discounted Cumulative Gain (NDCG)**

Handles graded relevance (e.g., 0=Irrelevant, 1=Relevant, 2=Highly Relevant). It rewards systems for putting highly relevant items at the top. One of the most widely used modern metrics.

- **Mean Reciprocal Rank (MRR)**

Measures how early the first relevant result appears. Used especially for question answering or fact retrieval.

P@k - Precision at k

1. Precision at k (P@k)

This measures the proportion of retrieved documents in the top k positions that are relevant. It is a "head-heavy" metric that ignores everything after rank k . 

$$P@k = \frac{\text{Number of relevant documents in top } k}{k}$$

- **Binary Relevance:** Documents are either relevant (1) or non-relevant (0).

<https://www.evidentlyai.com/ranking-metrics/precision-recall-at-k>

MAP - Mean Average Precision

2. Mean Average Precision (MAP)

MAP provides a single-figure measure of quality across recall levels. It is the mean of the **Average Precision (AP)** for a set of queries.

First, calculate **AP** for a single query:

$$AP = \frac{\sum_{k=1}^n (P@k \times rel_k)}{\text{Total number of relevant documents}}$$

(Where rel_k is 1 if the item at rank k is relevant, 0 otherwise.)

MAP is then the average of AP scores across all Q queries:

$$MAP = \frac{1}{|Q|} \sum_{q=1}^Q AP(q)$$

NDCG - Normalised Discounted Cumulative Gain

3. Normalized Discounted Cumulative Gain (NDCG)

NDCG is designed for **graded relevance** (e.g., 0 to 3). It rewards systems for putting highly relevant documents at the very top. 

1. **CG (Cumulative Gain)**: Sum of relevance scores.
2. **DCG (Discounted CG)**: Penalizes relevant documents found lower in the list. 

$$DCG_k = \sum_{i=1}^k \frac{rel_i}{\log_2(i + 1)}$$

3. **IDCG (Ideal DCG)**: The DCG score if the list were sorted perfectly by relevance. 
4. **NDCG**: Normalizes the score between 0 and 1.

$$NDCG_k = \frac{DCG_k}{IDCG_k}$$

MRR - Mean Reciprocal Rank

4. Mean Reciprocal Rank (MRR)

MRR is used when we only care about the **first** relevant document. It is the average of the Reciprocal Ranks (RR) across all queries. 

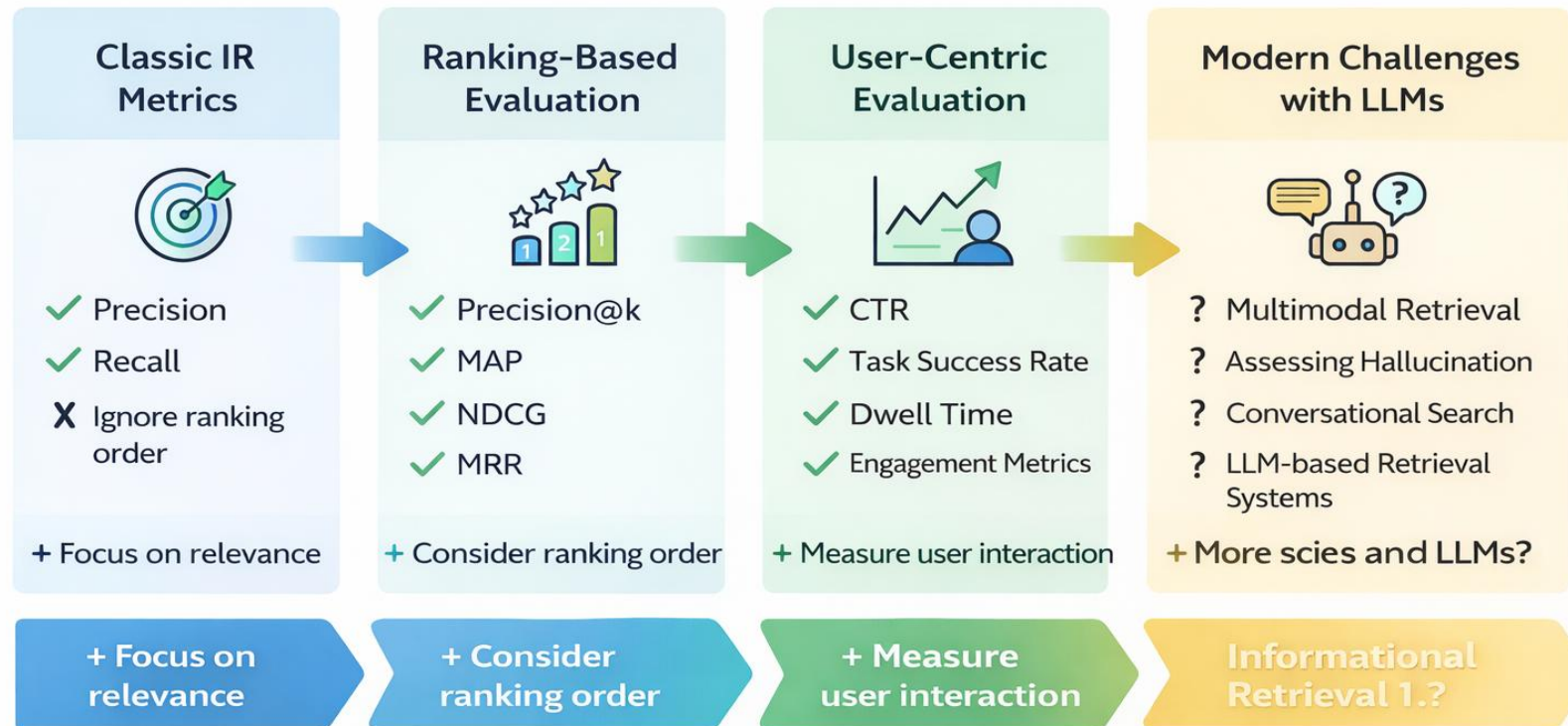
$$RR = \frac{1}{\text{rank of the first relevant document}}$$

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}$$

- If the first relevant document is at rank 1, $RR = 1$.
- If it is at rank 5, $RR = 0.2$.

<https://www.evidentlyai.com/ranking-metrics/mean-reciprocal-rank-mrr>

Evolution of IR evaluation



User-centric evaluation

Measure	Description	What it measures
Click-Through Rate (CTR)	Proportion of search results that users click after issuing a query.	Measures attractiveness and perceived relevance of results.
Dwell Time	Time a user spends on a clicked document before returning to the search results.	Longer dwell time often indicates higher user satisfaction or relevance.
Session Success	Whether a search session (possibly involving multiple queries) ends with a successful outcome.	Measures overall success of the search interaction, not just a single query.
Task Success Rate	Percentage of users who successfully complete an information task using the search system.	Captures effectiveness in supporting user goals.
Expected Reciprocal Rank (ERR)	Ranking metric that models the probability that a user finds relevant information at each rank position.	Captures user satisfaction based on graded relevance and stopping behaviour.
User Engagement	Degree of user interaction with the search system (clicks, scrolling, time spent, interactions).	Reflects interest, usability, and usefulness of the system.

ERR - Expected Reciprocal Rank

5. Expected Reciprocal Rank (ERR)

ERR is a more sophisticated version of MRR that accounts for graded relevance and the "cascade model"—the idea that a user might stop scanning the list after finding a relevant document. 

The probability that a user finds a document at rank i "satisfying" is defined as R_i :

$$R_i = \frac{2^{rel_i} - 1}{2^{max_rel}}$$

The ERR is the expected value of the reciprocal rank:

$$ERR = \sum_{i=1}^n \frac{1}{i} P(\text{user stops at rank } i)$$

Where $P(\text{stop at } i) = R_i \prod_{j=1}^{i-1} (1 - R_j)$.

Evaluation measures overview

Metric	Relevance type	Key focus
P@k	Binary	Precision in the top results.
MAP	Binary	Overall ranking quality across all relevant docs.
NDCG	Graded	Usefulness based on position and grade.
MRR	Binary	How "fast" the user finds the first answer.
ERR	Graded	Probability of user satisfaction at each rank.

Evaluation dimensions

Evaluation dimension	What it measures	Typical metrics / methods
Ranking Quality	How well relevant documents are ordered in the results list	MAP, NDCG, MRR, Precision@k
User Interaction	How users interact with search results	Click-through rate (CTR), dwell time, session success
Task Success	Whether users complete their information task	Task completion rate, time to task completion
Coverage / Diversity	Whether different aspects of a query are represented	α -NDCG, subtopic recall
Novelty	Whether new relevant information is presented	Expected Reciprocal Rank (ERR), novelty metrics
Efficiency / Scalability	System performance under large-scale conditions	Latency, indexing time, throughput
Robustness	Stability across queries and datasets	variance across queries, cross-dataset performance
Fairness / Bias	Whether ranking introduces systematic bias	exposure fairness metrics, demographic parity
Explainability / Transparency	Whether ranking decisions can be understood	explanation quality, interpretability measures

Statistical significance: is X better than Y?

Practical relevance of measurements: small ($\sim 2\%$) improvement in an indicator (MAP, ...) might be simply noise (random effects that occur by chance not by intrinsic competence of the system).

Run statistical tests to assure statistical significance:

- **Paired T-Test** or **Wilcoxon Signed-Rank Test**: essential for proving that System A is statistically better than System B over a set of queries.
- **Inter-Rater reliability**: using **Cohen's Kappa** to measure how much human judges agree on what is "relevant."

References

- <https://users.dcc.uchile.cl/~rbaeza/mir2ed/>
- <https://machinelearningmastery.com/>
- <https://www.evidentlyai.com/classification-metrics/explain-roc-curve>